

WEEK 14

Responsible bioinformatics and your scientific record

What can you share, claim and sign your name to?

By the end, you should be able to...



Distinguish pseudonymized data from a justified anonymity claim.



Identify when a data flow needs institutional privacy review.



Submit a reproducible portfolio with a precise AI-use disclosure.

Reconnect with the biology

Genetic and health information can be sensitive even when a name column has been removed.

Think silently -> explain to a neighbour -> check the measurement.

Removing names does not remove all risk

Pseudonymization

Anonymity claim

A code can replace a name while a linkage key remains elsewhere.

Genomic patterns and combinations of metadata can permit re-identification.

Anonymity is a contextual assessment, not a blanket promise about a file format.

Map the data flow before uploading

01

What data leave which institution?

>

02

Who receives them, where are they processed, and for what purpose?

>

03

Who can access them, for how long, and under what approval?

KVKK and GDPR are not interchangeable labels

KVKK treats genetic and health data within special categories.

Processing conditions and international-transfer requirements must both be considered.

GDPR applicability and obligations depend on the situation; seek institutional advice.

Consent is not a universal shortcut

- 1 A legitimate processing basis does not by itself solve every transfer or security obligation.
- 2 Minimize data, limit access and document purpose and retention.
- 3 Use the institution's approved environment for real sensitive data.

The classroom uses safer data by design

Synthetic examples do not represent real patients.

Public excerpts remain subject to source and license conditions.

Use course pseudonyms; do not embed student numbers in public notebooks.

Make an AI disclosure that can be checked

01

Name the tool and date or version where known.



02

Describe what it assisted with and which checks you performed.



03

State responsibility for the submitted result;
No AI used is acceptable.

Reproducibility has a practical minimum

- 1 Data provenance, code or steps, versions, parameters and independent unit.
- 2 A clean rerun plus a plain-language explanation of limitations.
- 3 A record of corrections is more useful than a claim of perfection.

Your portfolio is evidence of skills

01

D1: an experiment you can justify.

>

02

D2: a prediction you can evaluate honestly.

>

03

A model card, structure card and variant statement showing the limits of your claims.

Worked example: the cloud-upload decision

A colleague removes names from patient VCF files and proposes uploading them to a public chatbot for annotation.

Predict the result and one limitation before calculating.

Work through the reasoning

Genetic information remains; removing names does not establish anonymity.

The data flow and vendor processing can require privacy and transfer review.

Do not upload; consult the authorized institutional process and use synthetic examples for teaching.

Pause: make a defensible claim

Write a one-sentence AI disclosure for a notebook where a tool explained a Python error but did not choose the biological conclusion.

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

Example: I used [tool and date] to explain a Python error, checked the revised calculation with the stated test inputs, and independently wrote and take responsibility for the interpretation.

Today's practical

W14 studio: complete a data-flow decision card, a disclosure and a skills-ledger reflection. Bring your D1 notebook for the individual oral check.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 privacy
- 2 pseudonymization
- 3 transfer
- 4 disclosure
- 5 provenance

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: compare controlled-access and federated analysis designs without moving real patient data.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S20 Turkish Personal Data Protection Law No. 6698, official English translation with 2024 amendments.

S21 KVKK: international transfers, Article 9 amendments and safeguards.

S22 Regulation (EU) 2016/679 (GDPR), official EUR-Lex text.

S23 ICMJE recommendations: use of AI by authors.